Data Communication

Content 03: Data and Signals

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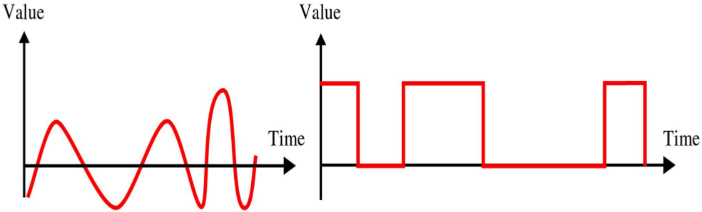
Data and Signals

- Information can be voice, image, numeric data, characters, code or picture
- Information must be into electromagnetic signals to be transmitted.

Analog and Digital Signals

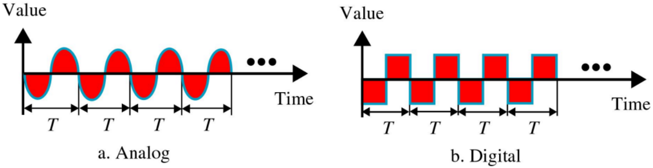
- The term **analog** data refers to continuous information. It has infinitely many levels of intensity over a period of time.
- **Digital** data refers to information that has discrete states. It can have only a limited number of defined values.



a. Analog signal b. Digital signal

Aperiodic and Periodic Signals

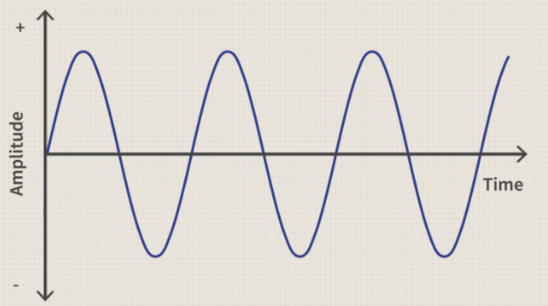
- Periodic signals consist of continuous cycles (a cycle = completion of one full pattern).
- Aperiodic signals change constantly without exhibiting a pattern or cycle that repeat over time.
- Example of periodic signals-



We commonly use periodic analog signals and aperiodic digital signals in data communication.

Periodic Analog Signals

- Periodic analog signals can be classified as **simple** or **composite**.
- A simple periodic analog signal, a sine wave, cannot be decomposed into simpler signals. The sine wave is the most fundamental form of a periodic analog signal.
- A composite periodic analog signal is composed of multiple sine waves.



Characteristics of Analog Signals

A sine wave can be fully described by three characteristics:

- **Amplitude** refers to the height of the signal. It is expressed in Amperes (A)/Volts (V)/Watts (W).
- **Period, Frequency**
  - **Period (T)** refers to the amount of time, in seconds, a signal needs to complete one cycle. It is expressed in seconds (s).
  - **Frequency (f)** refers to the number of periods a signal makes over the course of one second. It is expressed in Hertz (Hz).

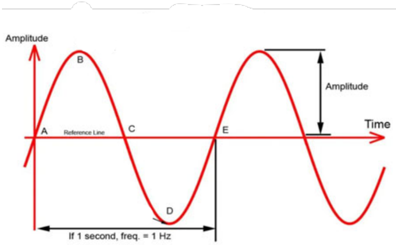
$$f = \frac{1}{T}$$

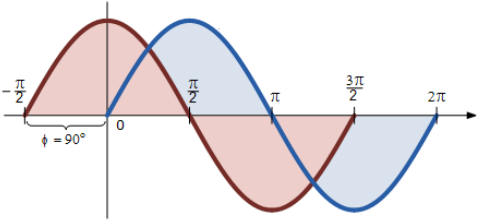
Characteristics of Analog Signals

- **Phase** describes the position of the waveform relative to time zero.

Sine waves can be expressed in the following formula:

$$f(t) = Asin(2\pi ft + \phi)$$



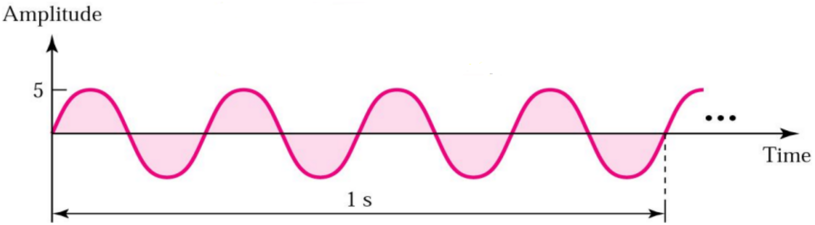


Examples of Analog Signals

Here,

$$A = 5 \quad f = 4 \quad \phi = 0$$

$$f(t) = 5sin(8\pi t + 0)$$

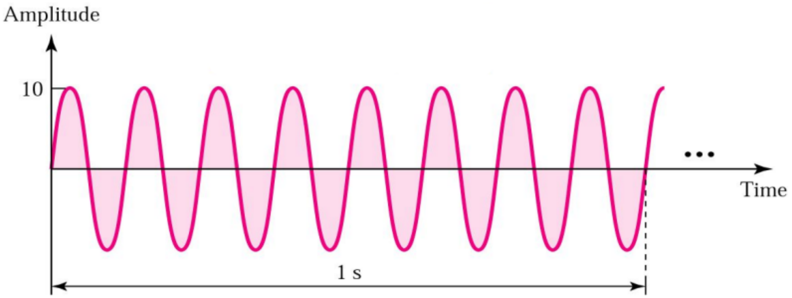


Examples of Analog Signals

Here,

$$A = 10 \quad f = 8 \quad \phi = 0$$

$$f(t) = 10sin(16\pi t + 0)$$

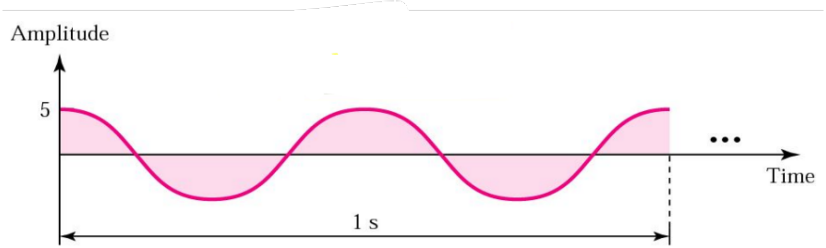


Examples of Analog Signals

Here,

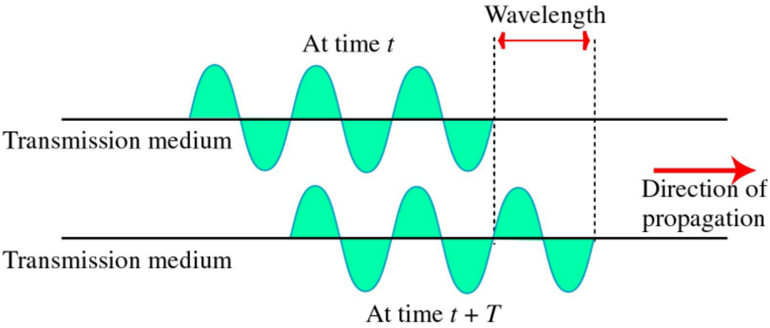
$$A = 5 \quad f = 2 \quad \phi = \frac{\pi}{2}$$

$$f(t) = 5sin(4\pi t + \frac{\pi}{2})$$



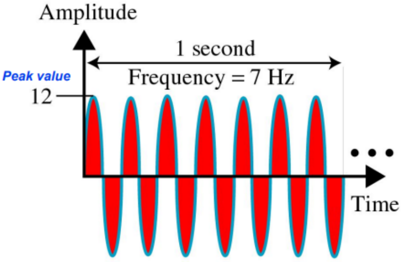
Wavelength

$$Wavelength, \lambda = \frac{c}{f} = \frac{Propagation \quad Speed}{Frequency}$$

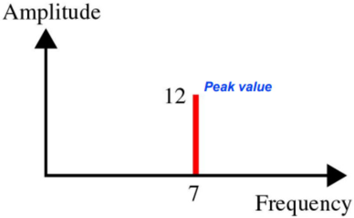


Time Domain vs Frequency Domain

- **Time Domain**: instantaneous amplitude with respect to time.
- **Frequency Domain**: maximum amplitude with respect to frequency.



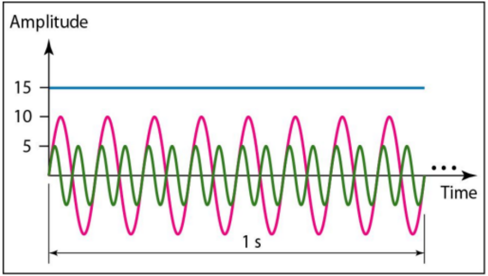
a. Time domain



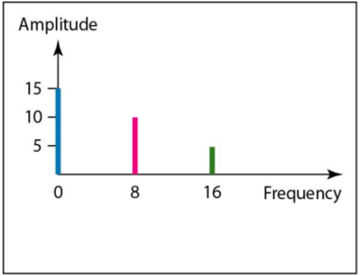
b. Frequency domain

Time Domain vs Frequency Domain

Periodic Analog Signal



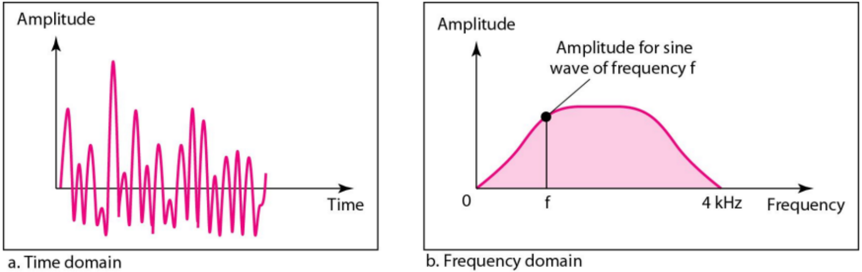
a. Time-domain representation of three sine waves with frequencies 0, 8, and 16



b. Frequency-domain representation of the same three signals

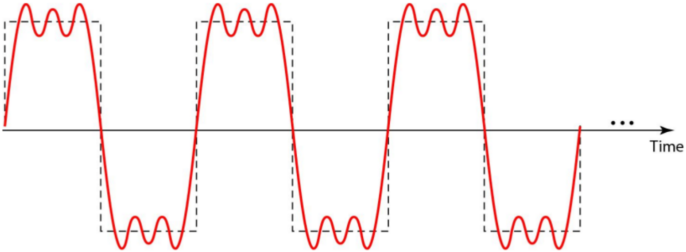
Time Domain vs Frequency Domain

Aperiodic Analog Signal

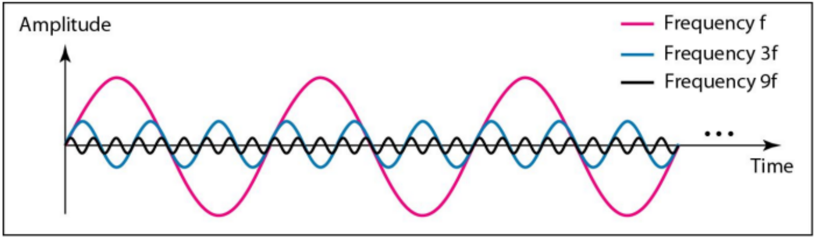


Composite Signals

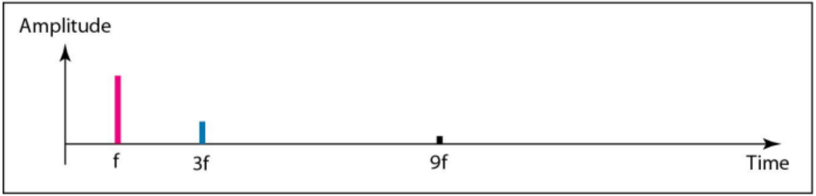
- According to Fourier analysis, any composite signal is a combination of simple sine waves with different frequencies, amplitudes, and phases.
- If the composite signal is periodic, the decomposition gives a series of signals with discrete frequencies; if the composite signal is non-periodic, the decomposition gives a combination of sine waves with continuous frequencies.



Composite Signal Decomposition



a. Time-domain decomposition of a composite signal



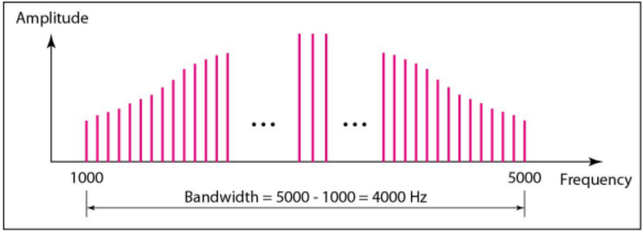
b. Frequency-domain decomposition of the composite signal

Bandwidth

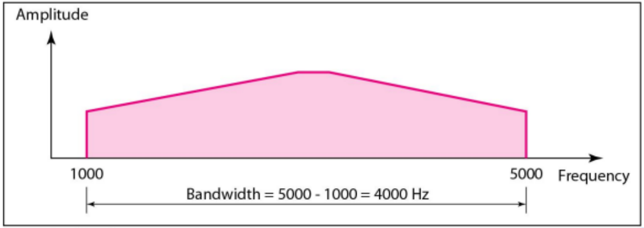
- The **frequency spectrum** of a signal is the combination of all sine wave signals that make the signal.
- The **bandwidth** of a composite signal is the difference between the highest and lowest frequencies contained in that signal.

$$Bandwidth, B = f_{highest} - f_{lowest}$$

Bandwidth



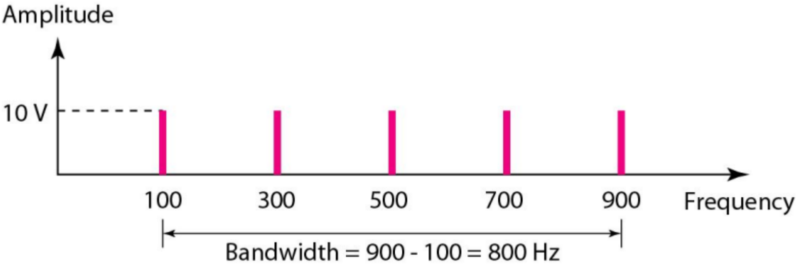
a. Bandwidth of a periodic signal



b. Bandwidth of a nonperiodic signal

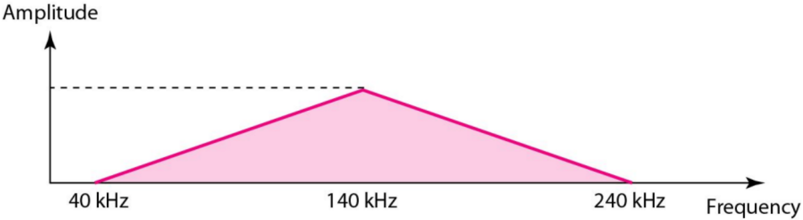
Bandwidth

If a periodic signal is decomposed into five sine waves with frequencies of 100, 300, 500, 700, and 900 Hz, what is its bandwidth? Draw the spectrum, assuming all components have a maximum amplitude of 10 V.



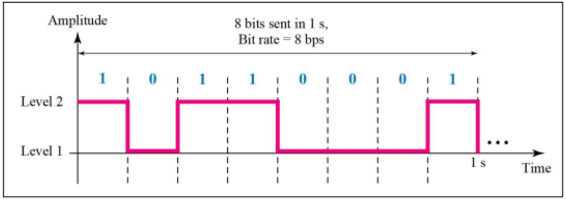
Bandwidth

A nonperiodic composite signal has a bandwidth of 200 kHz, with a middle frequency of 140 kHz and peak amplitude of 20 V. The two extreme frequencies have an amplitude of 0. Draw the frequency domain of the signal.

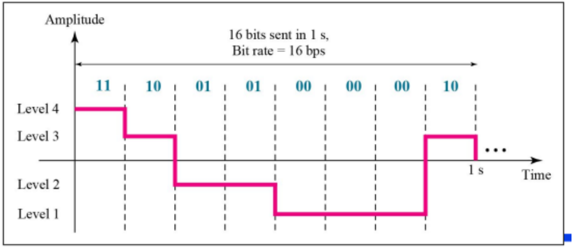


Digital Signals

A digital signal can have more than two levels.



a. A digital signal with two levels



b. A digital signal with four levels

Digital Signals

A digital signal has eight levels. How many bits are needed per level?

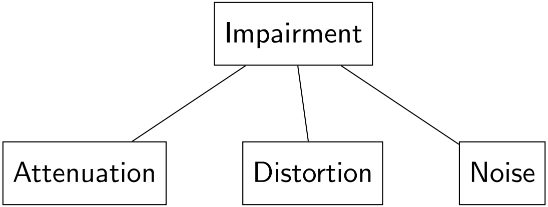
Formula:

$$\text{Numbers of levels required} = 2^r \text{ (for } r \text{ bits)}$$
$$\text{Number of bits per level} = log_2 L \text{ (for } L \text{ levels)}$$

For this example, number of bits per level $= log_2 8 = 3$
Hence, each signal level is represented by 3 bits.

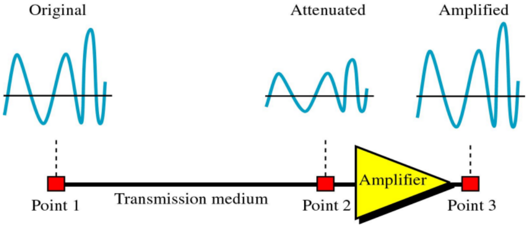
Transmission Impairment

- Transmission media are not perfect because of impairment in the signal sent through the medium.



Attenuation

- Attenuation means loss of energy
- When signal travels trough a medium, it losses some of it's energy
- To compensate for this loss, amplifiers are used to amplify the signal



Attenuation

$$dB = 10log_{10}\frac{p_{received}}{p_{sent}}$$

- If signal power is reduced to one-half:

$$p_2 = 0.5p_1$$

$$dB = 10log_{10}\frac{p_2}{p_1} = 10log_{10}\frac{0.5p_1}{p_1} = -3\ dB$$

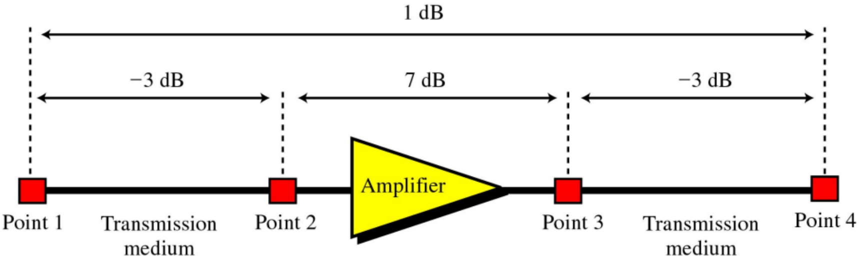
- If signal power is increased 10 times:

$$p_2 = 10p_1$$

$$dB = 10log_{10}\frac{p_2}{p_1} = 10log_{10}\frac{10p_1}{p_1} = 10\ dB$$

Attenuation

$$dB\ at\ point\ 4 = -3 + 7 - 3 = +1$$



Attenuation

Calculate the power of a signal with $dB = -30$

$$dB = 10\ log_{10}\ P = -30$$

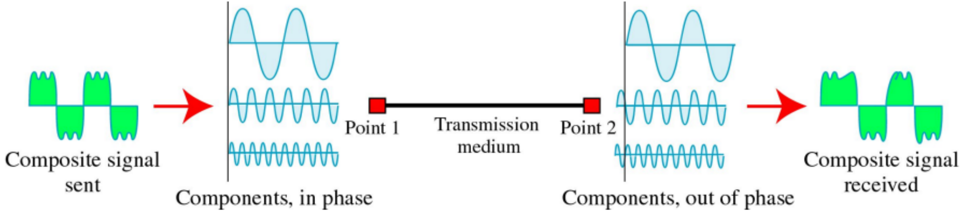
$$\Rightarrow log_{10}\ P = -3$$

$$\Rightarrow P = 10^{-3}$$

The received signal's power will be 10^{-3} times the sent signal's power.

Distortion

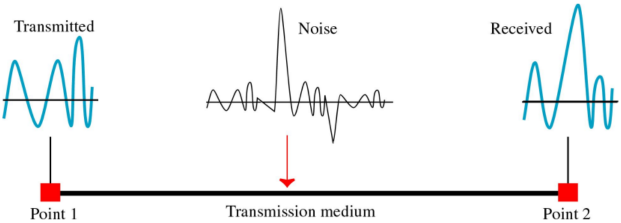
Distortion means that signal changes its form or shape.



Noise

Noise in data communication is any unwanted signal that disrupts or distorts the transmitted data. Noise types are-

- Thermal noise : random motion of electrons
- Induced noise : from sources such as motors, appliances
- Crosstalk : the effect of one wire on the other
- Impulse noise : a spike that comes from power lines, lightning



Signal to Noise Ratio

$$SNR = \frac{average\ signal\ power}{average\ noise\ power}$$

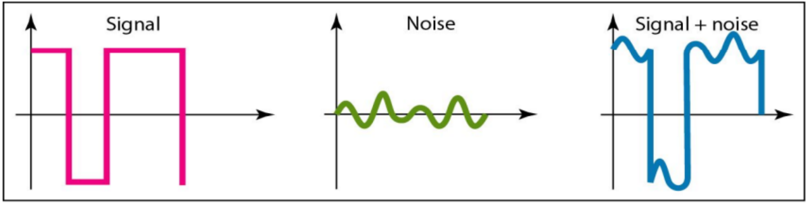
$$SNR_{dB} = 10\ log_{10}\ SNR$$

The power of a signal is 10 milliwatts and the power of the noise is 1 microwatts; what are the values of SNR and SNR_{dB} ?

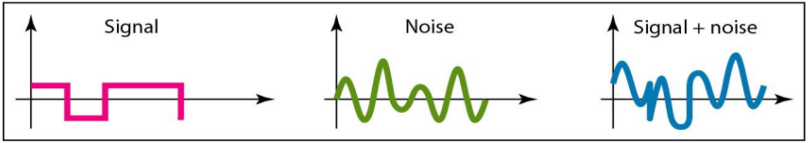
$$SNR = \frac{10 * 10^{-3}}{1 * 10^{-6}} = 10000$$

$$SNR_{dB} = 10\ log_{10}\ 10000 = 40$$

Signal to Noise Ratio



a. Large SNR



b. Small SNR

Transmission Impairments Summary

Impairment	Definition	Cause	Measurement	Mitigation
Attenuation	Loss of signal strength	Distance, resistance	dB formula: $10\log_{10}\frac{P_{sent}}{P_{received}}$	Amplifiers, repeaters
Distortion	Change in signal shape	Frequency-dependent delay	Oscilloscope observation	Equalizers
Noise	Unwanted interference	Thermal, crosstalk, impulse, intermodulation	Signal-to-Noise Ratio (SNR)	Shielding, filtering, error control

Data Rate Limits

Data rate is the speed at which data is transmitted over a communication channel, typically measured in bits per second (bps). Data rate depends on these three factors

1. The available bandwidth
2. The levels of signals
3. The quality of the channel (the level of the noise)

Data Rate Limits

For noiseless channel, Nyquist's formula is used.

$$Bit\ Rate = 2 * B * log_2\ L$$

L is the number of signal levels and B is the bandwidth of the channel.

Consider a noiseless channel with a bandwidth of 3000 Hz transmitting a signal with two signal levels. What will be the maximum bit rate?

$$Bit\ rate = 2 * 3000 * log_2\ 2 = 6000\ bps$$

Data Rate Limits

For noisy channel, Shannon's Capacity formula is used.

$$Capacity = B * log_2 \, (1 + SNR)$$

B is the bandwidth and SNR is the signal-to-noise of the channel.

A communication channel has a bandwidth of 500 kHz and a signal-to-noise ratio (SNR) of 30 dB. Calculate the maximum data rate that can be achieved.

$$SNR_{dB} = 10 \; log_{10} \; SNR$$

$$SNR = 10^{\frac{SNR_{dB}}{10}} = 10^{\frac{30}{10}} = 10^3 = 1000$$

$$C = B * log_2 \, (1 + SNR) = 500 * 10^3 * log_2(1 + 1000) = 4.985 \; Mbps$$

Data Rate Limits

We have a channel with a 1-MHz bandwidth. The SNR for this channel is 63. What are the appropriate bit rate and signal level?

$$C = B * log_2\ (1 + SNR) = 1 * 10^6 * log_2(1 + 63) = 6\ Mbps$$

Finding the required signal levels if the channel were noiseless,

$$Bit\ Rate = 2 * B * log_2\ L$$

$$L = 2^{\frac{C}{2*B}} = 2^{\frac{6*10^6}{2*1*10^6}} = 8$$

Shannon's formula gives the max data rate with noise, while Nyquist's formula finds required signal levels in a noiseless channel. In practice, We use **Shannon first for capacity**, **then Nyquist for signal levels** if needed.

Performance

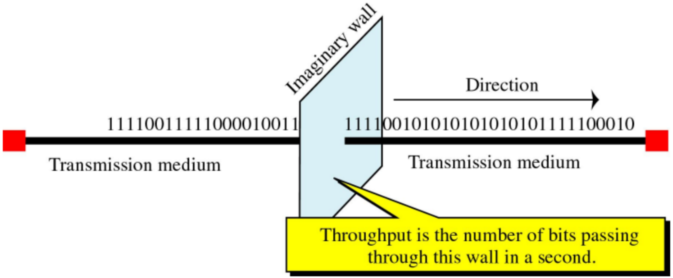
Performance refers to how efficiently a system transmits data, typically measured by throughput and latency.

Note: Bandwidth refers to either the frequency range of a signal (Hz) or the data transmission speed (bps).

Throughput

Throughput is the measurement of how fast data can pass through a point.

$$\text{Throughput} = \frac{\text{Total data transferred}}{\text{Total time taken}}$$



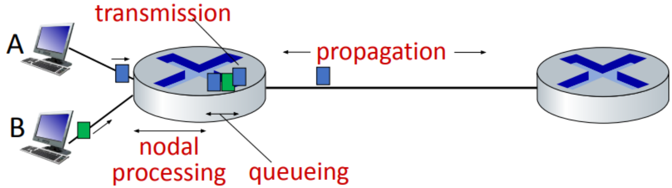
Throughput

A network with bandwidth of 10 Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of this network?

$$\text{Throughput} = \frac{\text{Total data transferred}}{\text{Total time taken}} = \frac{12000 * 10000}{60} = 2\ Mbps$$

Latency

The latency or delay defines how long it takes for an entire message to completely arrive at the destination from the time the first bit is sent out from the source.



$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

Practice Problems

- Theoretical:
  Data Communication and Networking (4th Ed) - Behrouz A. Forouzan
  Chapter 03 - 7, 8
- Mathematical:
  Data Communication and Networking (4th Ed) - Behrouz A. Forouzan
  Chapter 03 - 19, 27, 30, 31, 40, 42, 45, 48

References

- Data Communication and Networking (4th Ed) - Behrouz A. Forouzan
  Chapter 03 - 3.1, 3.2, 3.3, 3.4, 3.5, 3.6
- Computer Networking: A Top Down Approach (6th Ed) - Jim Kurose
  Chapter 01 - 1.3, 1.4

Thank You